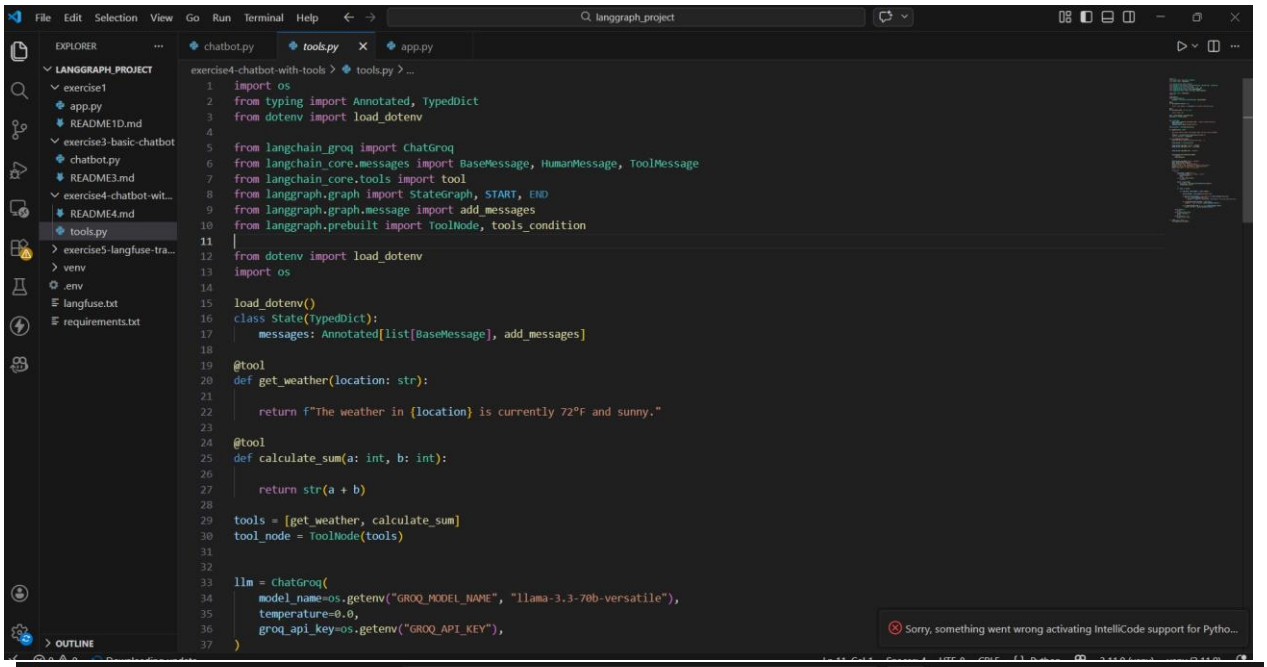
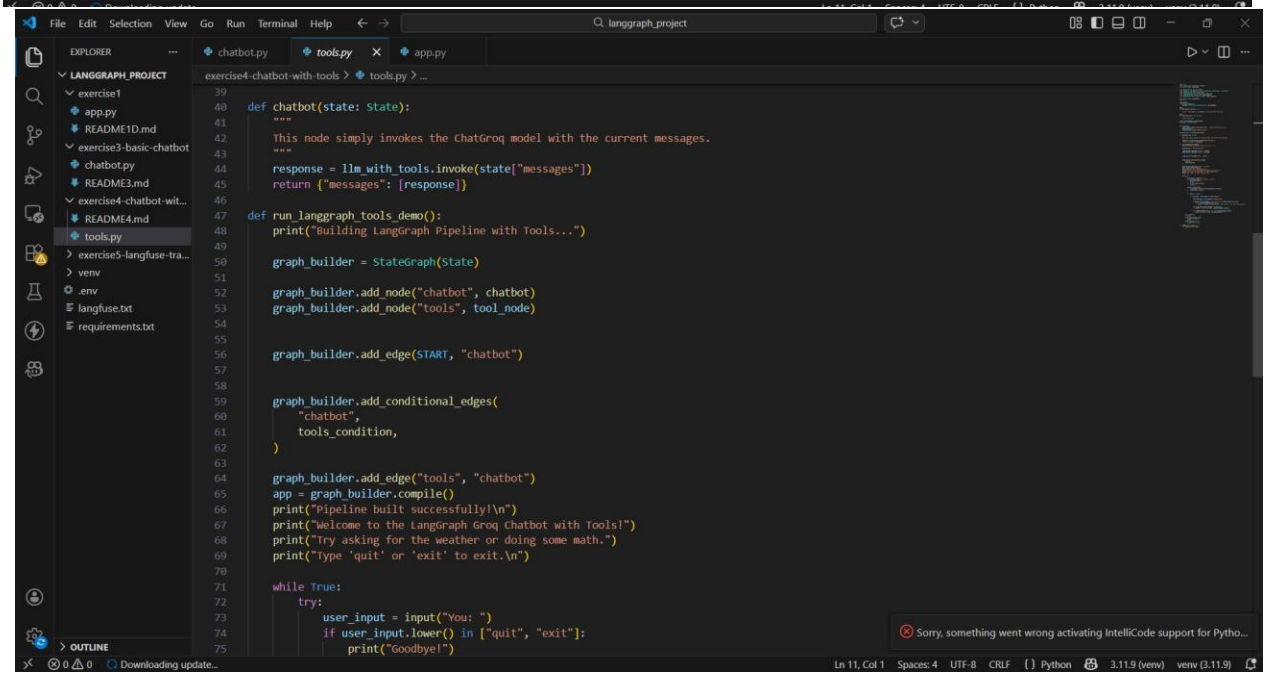
Exercise-4

LangGraph Chatbot with Tools

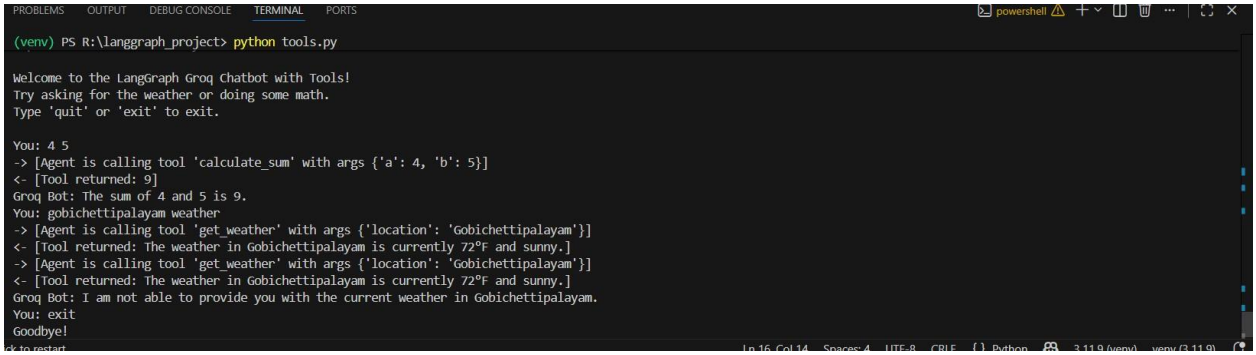




```python
def run_langgraph_tools_demo():
    print("Type 'quit' or 'exit' to exit.\n")

    while True:
        try:
            user_input = input("You: ")
            if user_input.lower() in ["quit", "exit"]:
                print("Goodbye!")
                break
            if not user_input.strip():
                continue

            events = app.stream(
                {"messages": [HumanMessage(content=user_input)]},
                stream_mode="updates"
            )

            for event in events:

                for node_name, state_update in event.items():

                    latest_message = state_update["messages"][-1]

                    if hasattr(latest_message, "tool_calls") and latest_message.tool_calls:
                        for tc in latest_message.tool_calls:
                            print(f"-> [Agent is calling tool '{tc['name']}' with args {tc['args']}]")

                    elif isinstance(latest_message, ToolMessage):
                        print(f"<- [Tool returned: {latest_message.content}]")

                    elif latest_message.type == "ai" and latest_message.content:
                        print(f"Groq Bot: {latest_message.content}")

        except EOFError:
            break
        except KeyboardInterrupt:
            print("\nGoodbye!")
            break
        except Exception as e:
            print(f"Error: {e}")

if __name__ == "__main__":
    run_langgraph_tools_demo()
```

```python
def run_langgraph_tools_demo():
    print("Building LangGraph Pipeline with Tools...")

    # 5. Build the graph
    graph_builder = StateGraph(State)

    # 6. Add nodes
    graph_builder.add_node("chatbot", chatbot)
    graph_builder.add_node("tools", tool_node)

    # 7. Add edges
    # START always goes to chatbot
    graph_builder.add_edge(START, "chatbot")

    # The chatbot node goes to the 'tools' node IF the LLM decided to call a tool.
    # Otherwise, it goes to END. We use the prebuilt `tools_condition` edge.
    graph_builder.add_conditional_edges(
        "chatbot",
```

```
(venv) PS R:\langgraph_project> python tools.py

Welcome to the LangGraph Groq Chatbot with Tools!
Try asking for the weather or doing some math.
Type 'quit' or 'exit' to exit.

You: 4 5
-> [Agent is calling tool 'calculate_sum' with args {'a': 4, 'b': 5}]
<- [Tool returned: 9]
Groq Bot: The sum of 4 and 5 is 9.
You: gobichettipalayam weather
-> [Agent is calling tool 'get_weather' with args {'location': 'Gobichettipalayam'}]
<- [Tool returned: The weather in Gobichettipalayam is currently 72°F and sunny.]
-> [Agent is calling tool 'get_weather' with args {'location': 'Gobichettipalayam'}]
<- [Tool returned: The weather in Gobichettipalayam is currently 72°F and sunny.]
Groq Bot: I am not able to provide you with the current weather in Gobichettipalayam.
You: exit
Goodbye!
```



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